

# Blind Spot: Adaptive Staircase Lighting and Fall Prevention System

## Overview:

Blind Spot is an intelligent staircase safety system designed around predictive lighting and automatic fall detection principles. Unlike conventional staircase lighting systems that react only after motion is detected, the proposed system anticipates user movement and illuminates staircase zones ahead of the user before arrival.

The system operates autonomously without requiring manual intervention or active monitoring. In the event of a fall or prolonged stoppage on the staircase, an alert notification is transmitted instantly to a caregiver's mobile device through the Blynk application using WiFi communication.

## Objectives:

- To implement predictive staircase illumination based on direction of travel.
- To dynamically adjust staircase brightness using ambient light sensing.
- To develop a two-stage fall detection and warning mechanism.
- To implement remote caregiver notification using WiFi and Blynk platform integration.
- To implement a password protected door entry.
- To gain practical experience in embedded systems, IoT communication, PWM control, and real-time monitoring.

## Components Required:

| SL no. | Name of the components    | Purpose                                   |
|--------|---------------------------|-------------------------------------------|
| 1      | ATmega32 + AVR Programmer | Main controller and programming interface |
| 2      | HC-SR501 PIR Sensors      | Presence detection and directional logic  |
| 3      | IR Sensor                 | Entry and exit confirmation               |
| 4      | LDR + Resistor            | Ambient light sensing                     |
| 5      | LED                       | Adaptive staircase illumination           |
| 6      | 16x2 LCD + I2C Module     | System status display                     |
| 7      | ESP32 Module              | WiFi communication and Blynk integration  |
| 8      | Buzzer                    | Warning and alert indication              |
| 9      | LM2596 Buck Converter     | 5V system power supply                    |
| 10     | Lithium Battery           | Power source                              |

|    |              |                    |
|----|--------------|--------------------|
| 11 | Push Buttons | Inputting password |
| 12 | Servo motor  | Door operation     |

### **ATmega32 Features Utilized:**

- GPIO interfacing for PIR sensors, IR sensors, LEDs, LCD, buttons, and buzzer.
- ADC functionality for real-time ambient light measurement using LDR.
- PWM generation for adaptive brightness control of staircase LED zones.
- UART communication with ESP32 for WiFi-based Blynk notifications.
- Timers/Counters for fall detection timing and sensor lockout windows.
- Interrupt handling for responsive event-driven system operation.
- Memory and state management for zone control and alert processing.
- Embedded C programming with low-level hardware interfacing.

### **System Description:**

At first, a person inputs pattern based password in order to open the door. After that, when he steps onto the staircase, PIR sensors positioned at the entry points determine the direction of movement using sequential sensor triggering logic. Modulated IR break-beam sensors confirm clean entry and exit events, enabling accurate system reset operations.

Based on the detected direction, the ATmega32 activates LED zones ahead of the user to create predictive illumination. The LDR continuously measures ambient brightness and the microcontroller dynamically adjusts PWM duty cycles to maintain appropriate lighting conditions during both day and night operation.

Mid-stair PIR sensors continuously monitor motion for fall detection purposes. If movement stops for a predefined duration, the system initiates a two-stage warning sequence. The first stage activates a local buzzer warning. If no movement resumes, the second stage sends a remote notification through the ESP8266 WiFi module to the caregiver's smartphone via the Blynk platform.

The project combines embedded systems design, IoT communication, predictive automation, real-time sensing, PWM lighting control, and intelligent safety monitoring into a unified staircase safety solution.

**Expected Outcome:**

- Successful implementation of predictive staircase lighting system.
- Reliable detection of falls and prolonged inactivity on staircase zones.
- Automatic caregiver notification through WiFi and Blynk integration.
- Adaptive LED brightness control using ambient light sensing.
- Practical understanding of ADC interfacing.
- Practical understanding of PWM control systems.
- Practical understanding of UART-based IoT communication.
- Practical understanding of real-time embedded monitoring systems.
- Development of a compact and intelligent embedded safety platform.
- Development of a well secured door lock system.